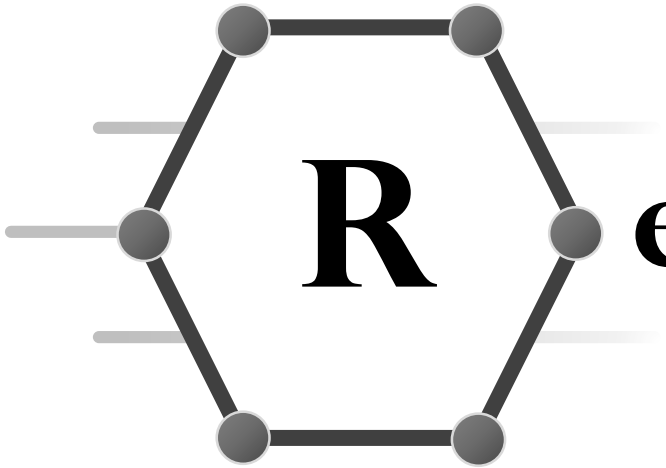


Resilience

Magnetic field resilient microwave single photon detector based on van der Waals Josephson junctions

Alessandro D'Elia, INFN Laboratori Nazionali di Frascati

Why and How

Resilience

Why

- Single photon detector in the microwave range based on Josephson effect is crucial for dark matter search (axions) and in general for quantum sensing
- However this technology as a main limitation: strong magnetic field kills superconductivity, thus Josephson junctions

How

- Fabrication of Josephson junctions (JJ) using van der Waals (vdW) materials
- Using $NbSe_2$ as superconductor transfers its magnetic field resiliency to the JJ and thus to the microwave single photon detector

Outline

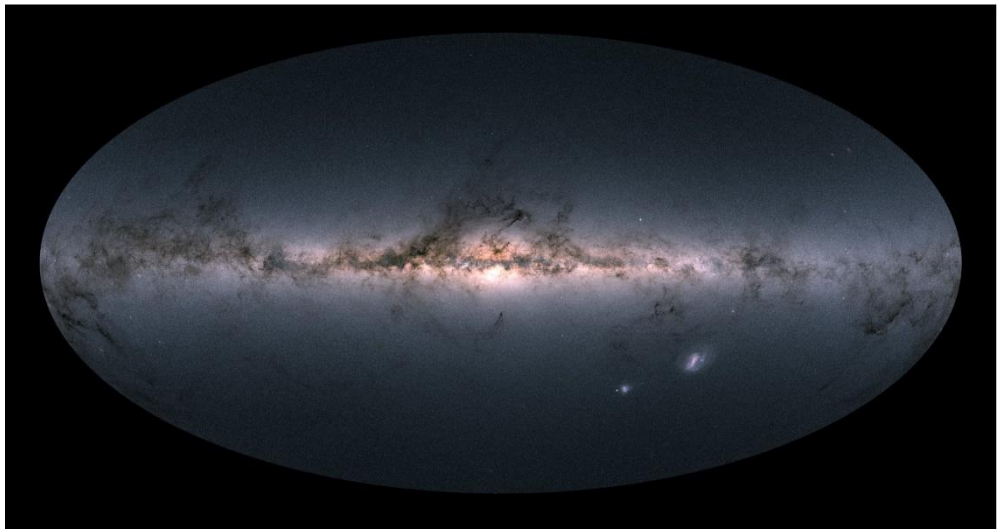
Background

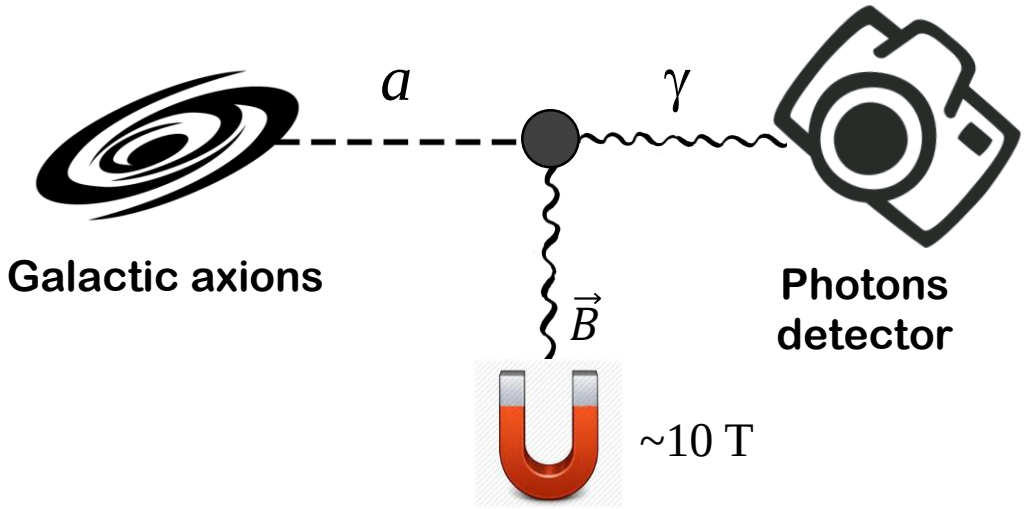
- Axions and the necessity for single microwave photon detector
- Josephson junctions
- Current limitations of JJ

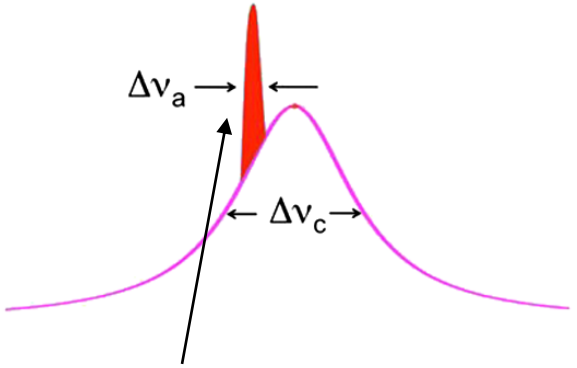
Proposal

- vdW materials as platform for quantum technologies
- $NbSe_2$ junctions to overcome the limitations
- Timetable, team composition and budget required

Axions as dark matter

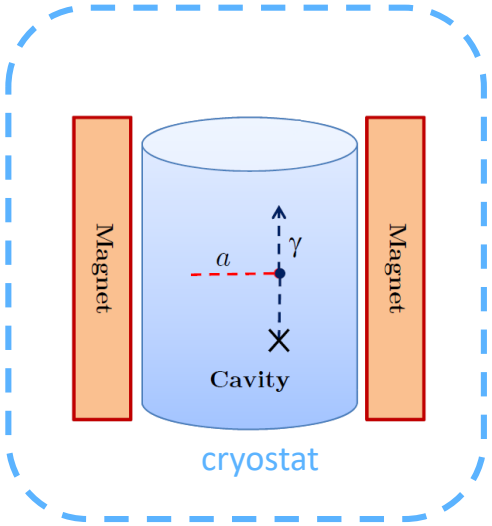




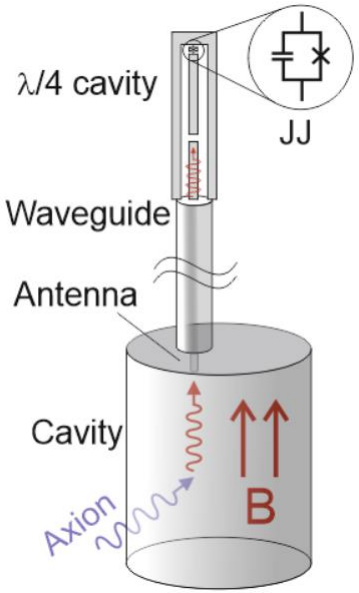


Localized power excess ~10^{-24} W
@ 10 GHz

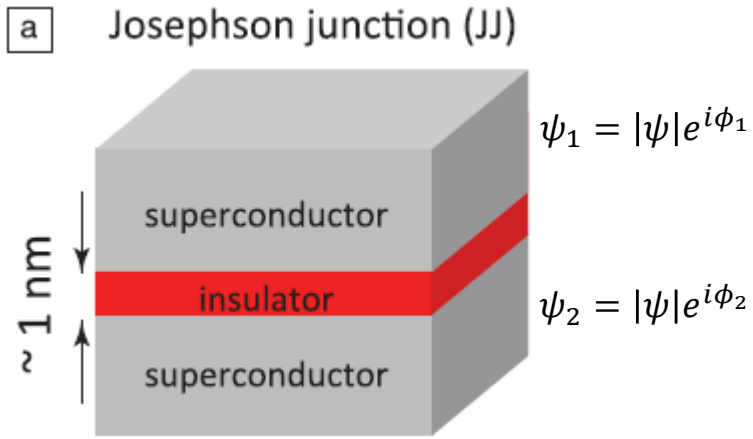
Haloscope



JJ as microwave single photon detector



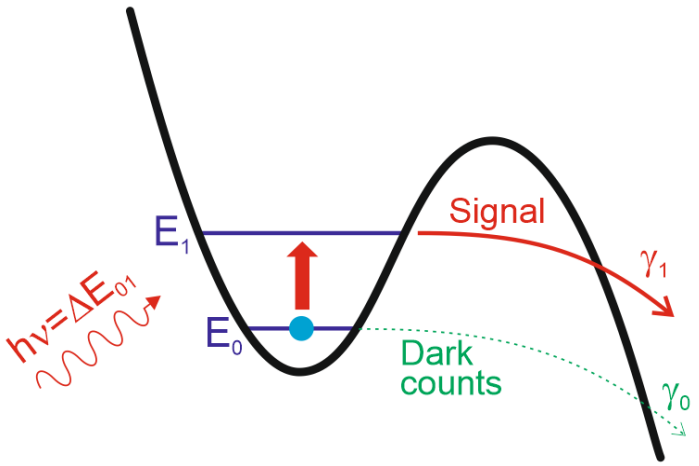
Josephson junction



$$\phi_1\text{-}\phi_2 = \varphi$$

$$I = I_c \sin\varphi$$

$$\frac{d\varphi}{dt} = \frac{2e}{\hbar}V$$



The JJ can be modelled as a fictitious phase particle trapped in a potential well

JJ as photon detector

The absorption of a photon, excites the phase particle, increasing the probability of tunnelling trough the barrier

The JJ switch from superconductive to normal

A finite voltage appears and can be measured

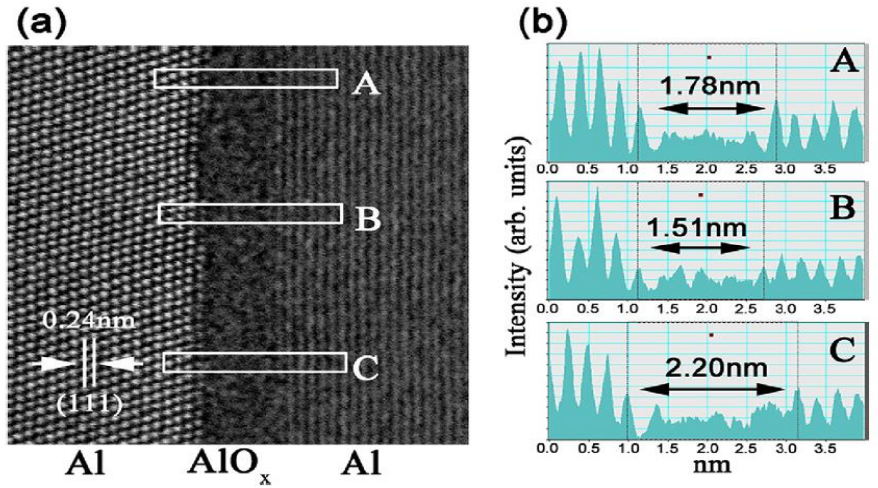
W. D. Oliver, P.B. Welander, MRS Bulletin, 2013

What's the problem?

Strong magnetic filed (~10 T) required for axion conversion kills the JJ

- Al is the superconductor most used to fabricate JJ: <u>critical field ~ 0,01T</u>
- Nb is the second most used: <u>critical field ~0,8T</u>

Standard fabrication methods, results in inhomogeneous insulating layer



Inhomogeneity in the oxide layer thickness can strongly hinder the junction performance

L. J. Zeng et al J. Phys. D: Appl. Phys. 2015

van der Waals Materials

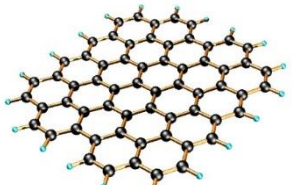
Resilience

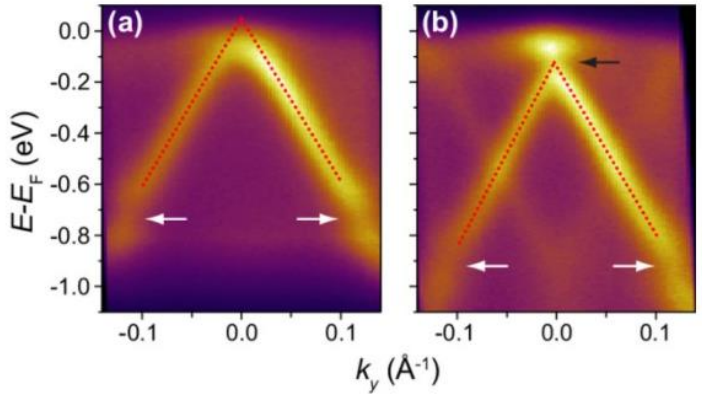
van der Waals (vdW) materials are crystals composed of a stack of atomically thin layers glued together by vdW forces.

They can be exfoliated (synthesized) down to monolayer thickness, pushing the quantum confinement to its limit.

New properties respect to the bulk counterpart.

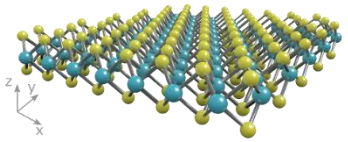
Graphene

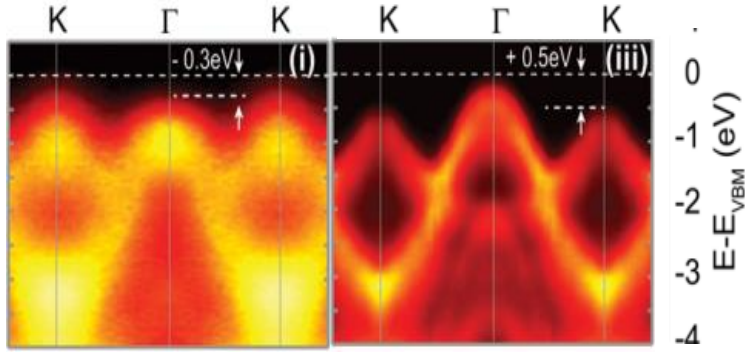




Marin Petrović et al. Phys. Rev. B, 2017

MoS_2

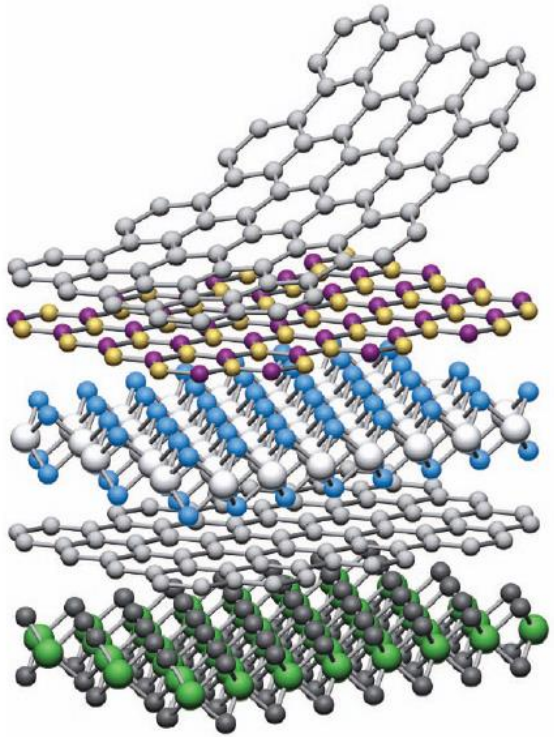


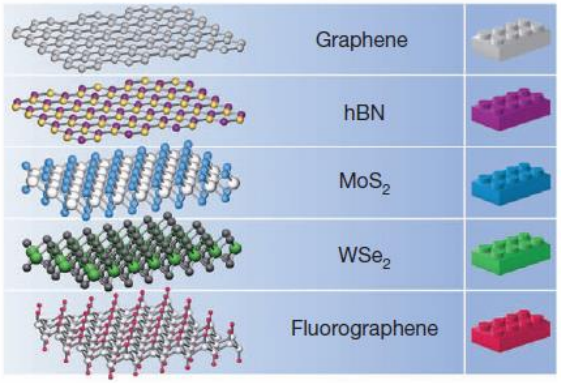


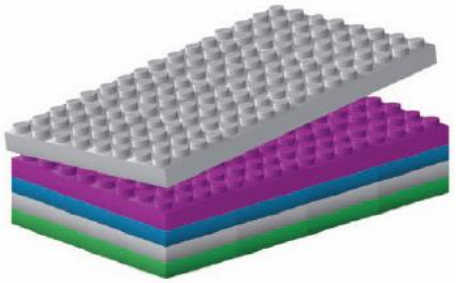
H. Yuan et al, Nano Lett. 2016

van der Waals Materials

- New functional properties (mass-less carrier, improved photon sensitivity, Ising superconductivity etc.)
- Flat surfaces and clean interfaces => Reduced amount of defects
- vdW materials can be assembled like LEGO to form heterostructures







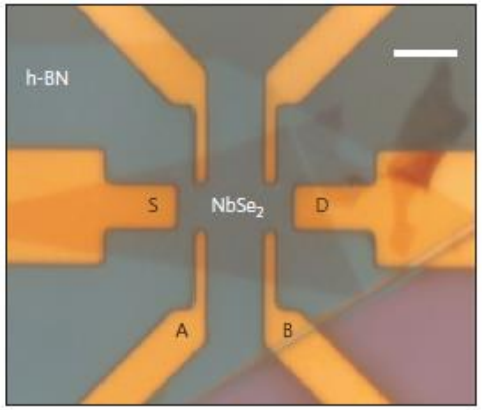
Integration
With current JJ fabrication technologies

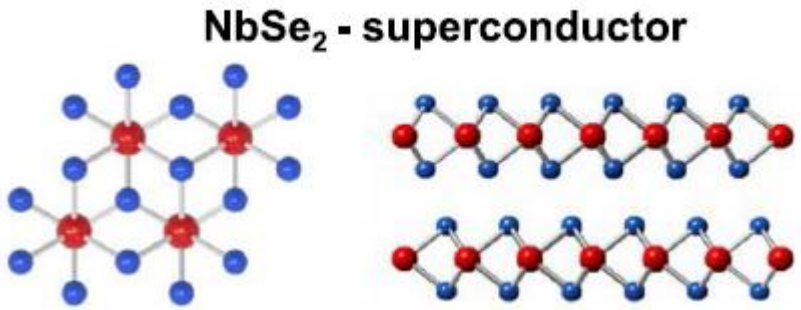
All vdW JJ

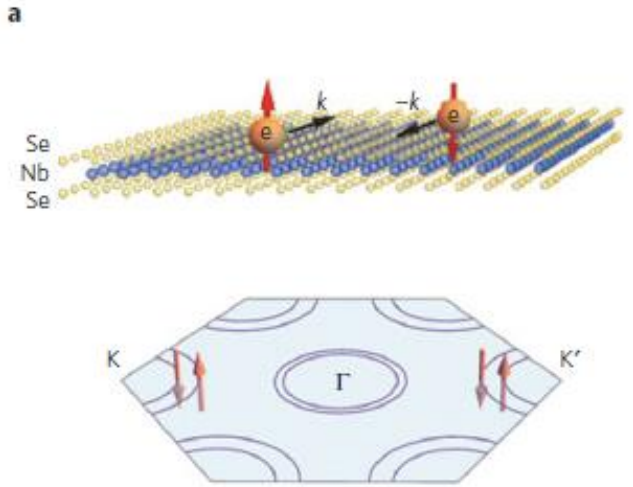
Geim, A., Grigorieva, I. *Nature* **499,** 419–425 (2013)

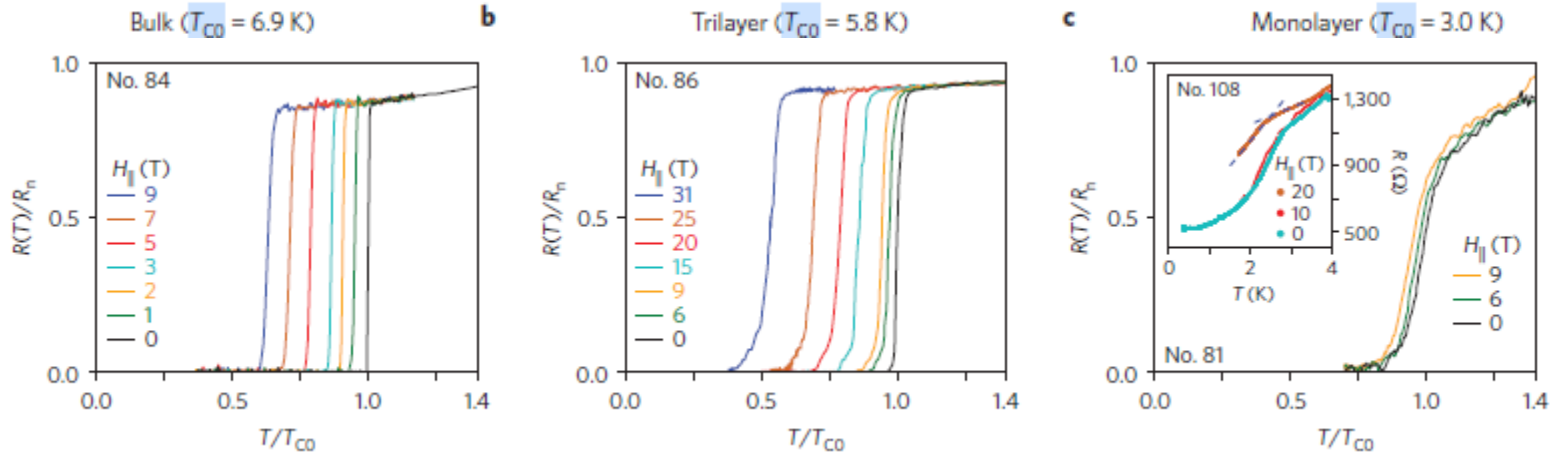
Ising superconductivity in $NbSe_2$

Resilience









Spin of Cooper pair is locked out of plane by a combination of strong spin orbit coupling and quantum confinement

Magnetic field resistance up to ~30T !!!

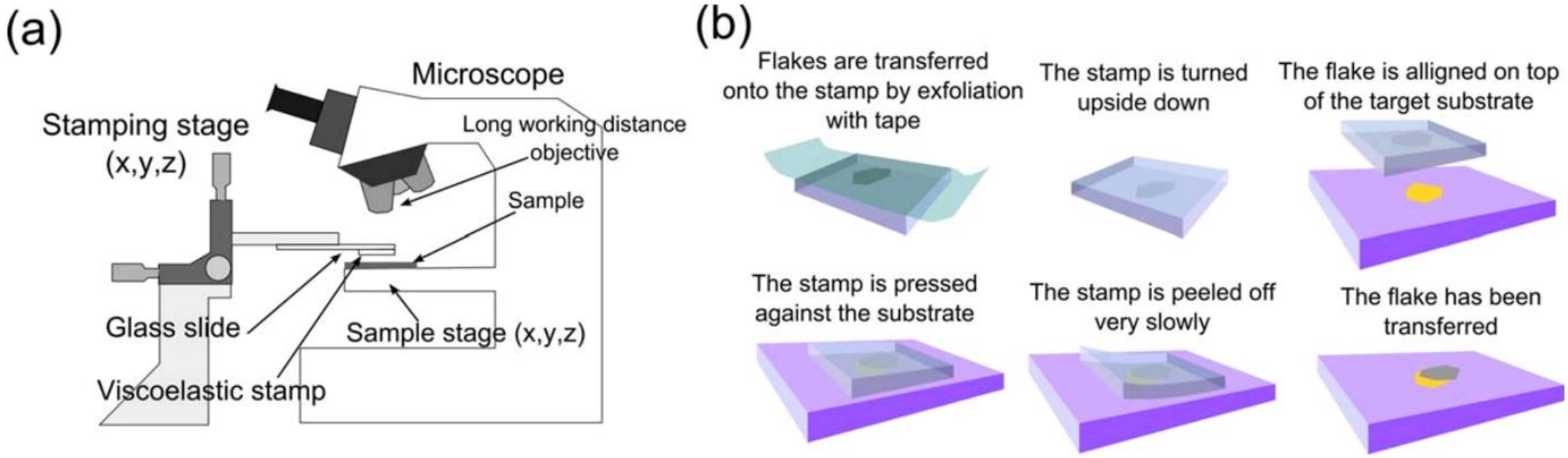
Xi, X., et al. Nature Phys (2016).

Resilience goals

Assembly of a $NbSe_2$ based magnetic field resilient Josephson junction

Test the device performance as single photon detector in presence of strong magnetic field ~10 T

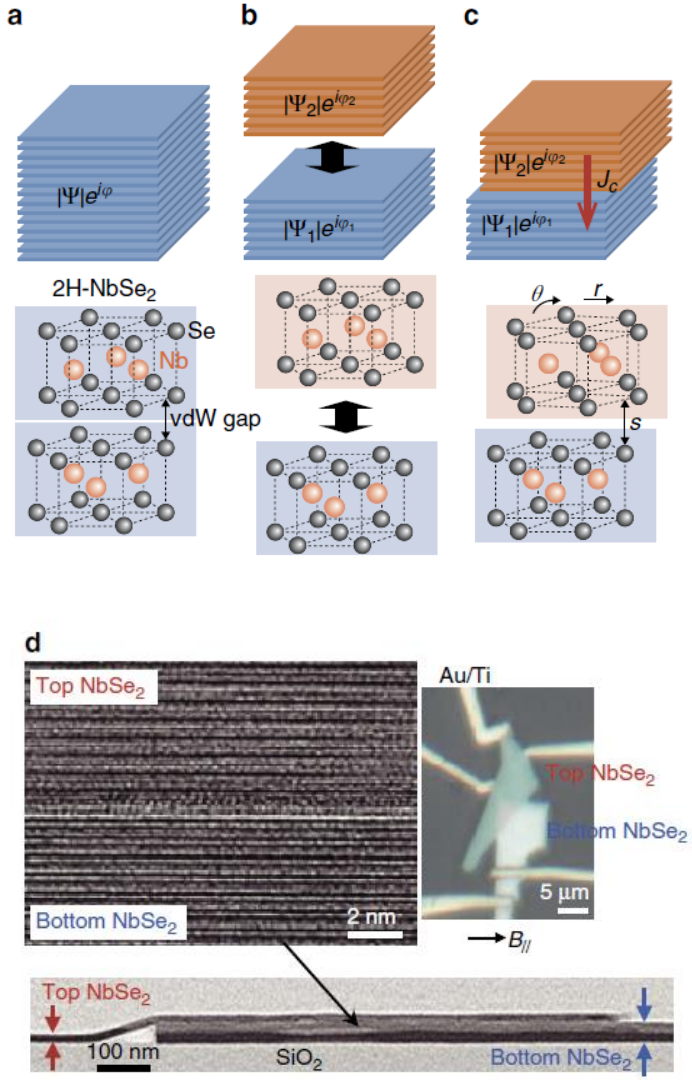
All vdW JJ assembly



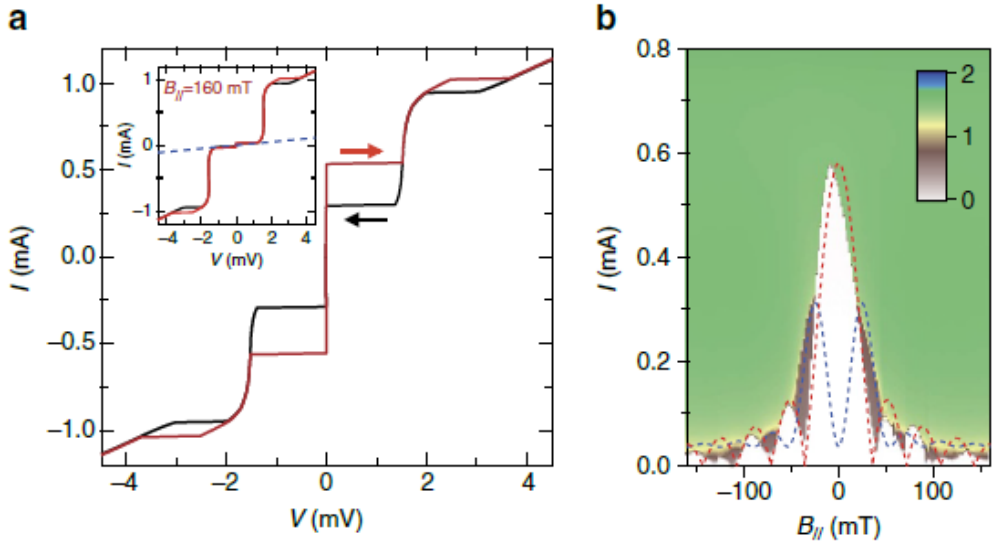
Andres Castellanos-Gomez *et al* 2014 *2D Mater.* **1** 011002

$NbSe_2$ based JJ are possible?

Resilience



Decoupling the superconductors wave functions exploiting the twisting angle rotation

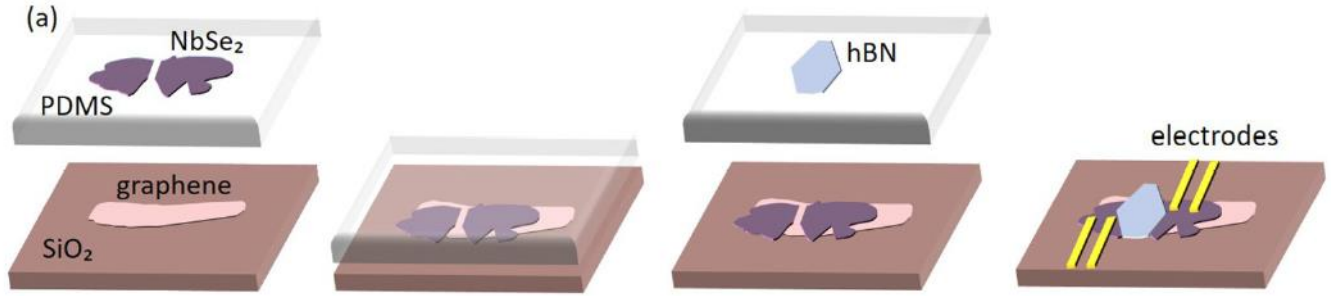


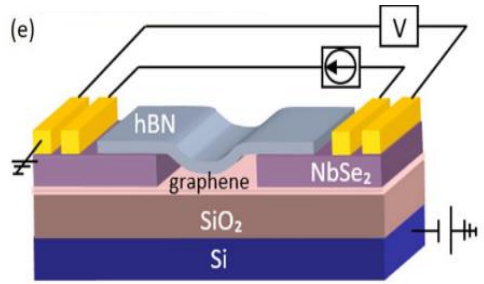
Yabuki et al. *Nat Commun*, 2016

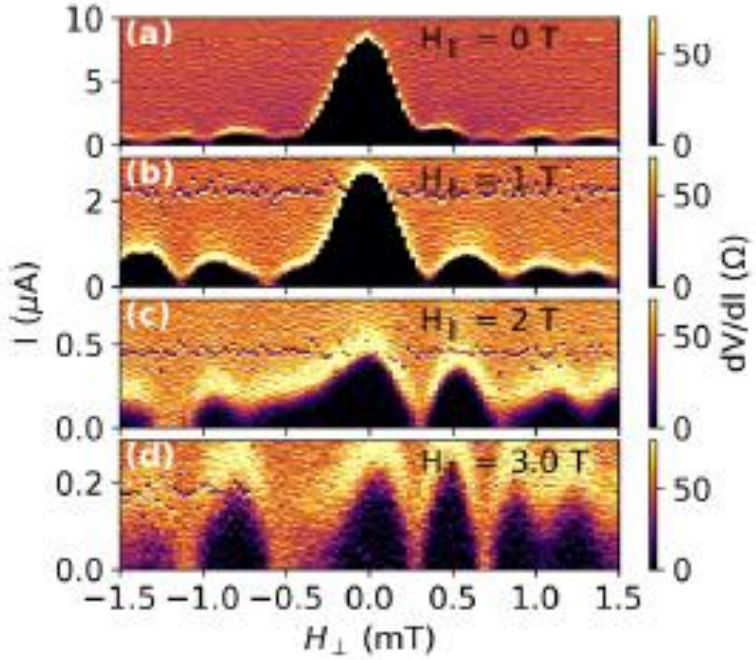
$NbSe_2$ based JJ are possible?

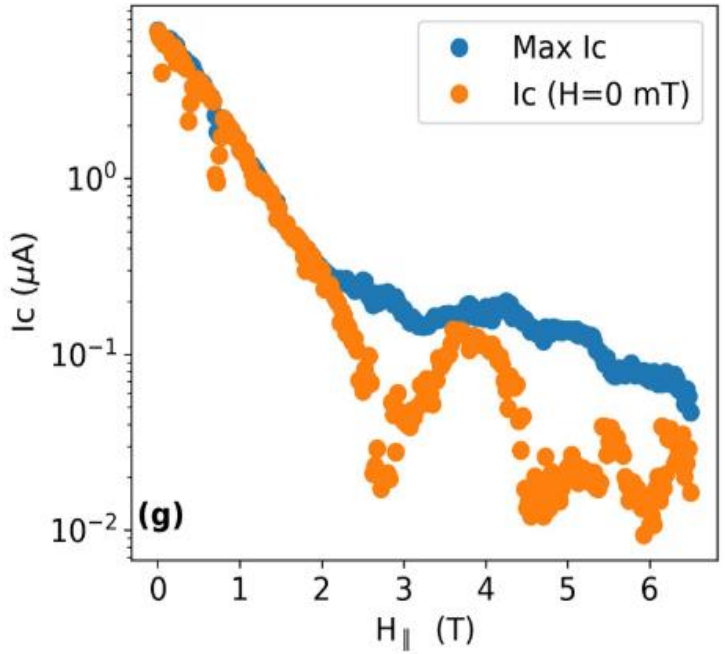
Resilience

T. Dvir et al. PRB, 2021









Optimization of the JJ resilience as a function of flakes thickness is still missing!

Resilience challenges



- **Find the assembly parameters (thickness of the SC layers and of the insulating layer), that yield the best magnetic field resiliency.**

HOW?

- **Determine the flakes thickness quickly using Raman spectroscopy**

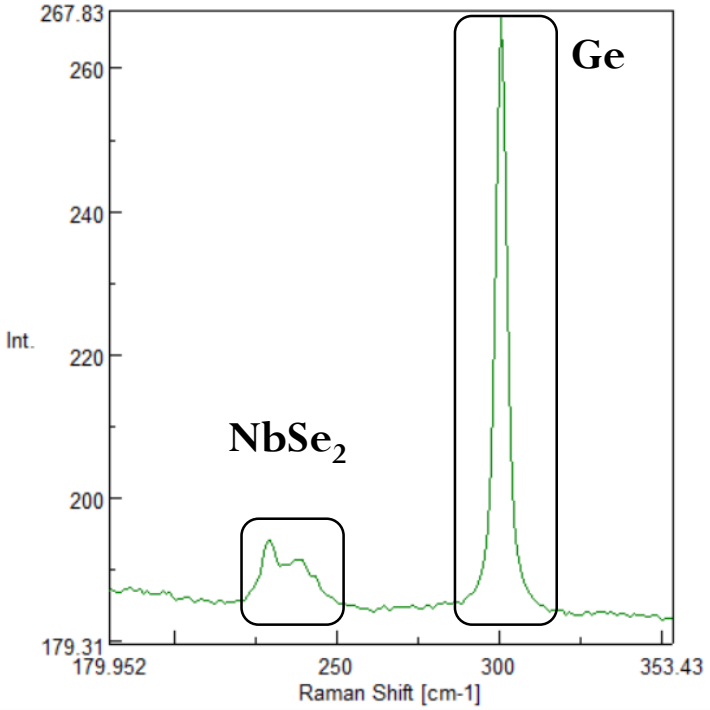
- **Fabricate JJ with deterministic number of layers of the superconductors and insulating materials and test their magnetic field resiliency.**

Resilience challenges

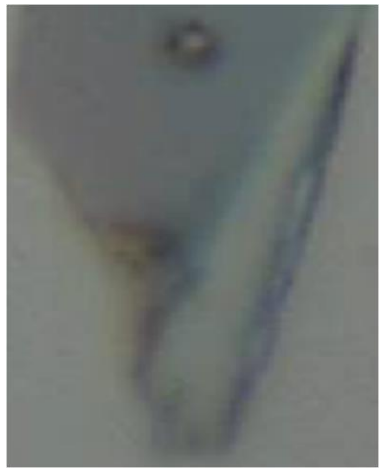
Discriminate between $NbSe_2$ flakes of different thickness

- **Measure the flakes thickness with AFM**
- **Calibrate the flakes' Raman spectra with the measured AFM thickness obtaining an Intensity vs thickness curve**

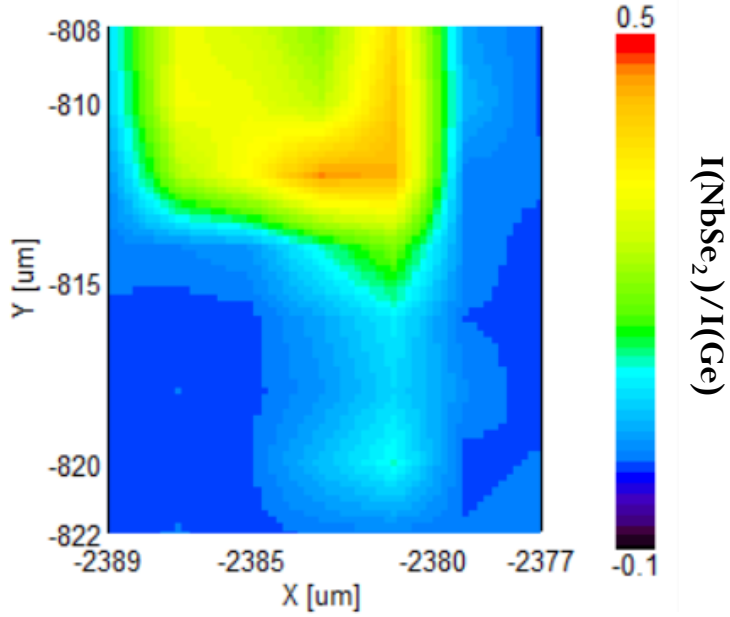
Raman spectrometer available @LNF



Intensity ratio of the $NbSe_2$ feature and Ge substrate feature depends on the $NbSe_2$ flake thickness



optical image of a $NbSe_2$ flake



Raman map of the flake

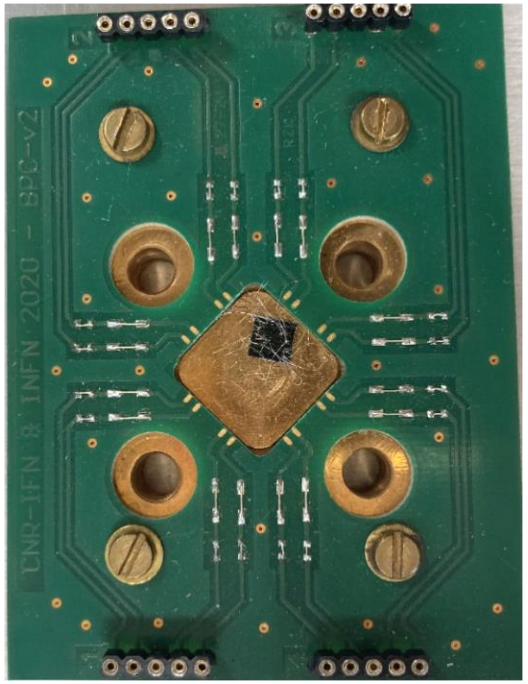
Resilience challenges

Maximizing the magnetic field resilience, i.e. optimize the numbers of $NbSe_2$ layers and thickness of insulating barrier

- Assembly of a batch of JJ with different thickness of the superconducting and insulating layers

- Test the I-V curves of these samples (@ COLD-LAB) down to L-He temperature and in presence of magnetic field up to 8T





Resilience challenges

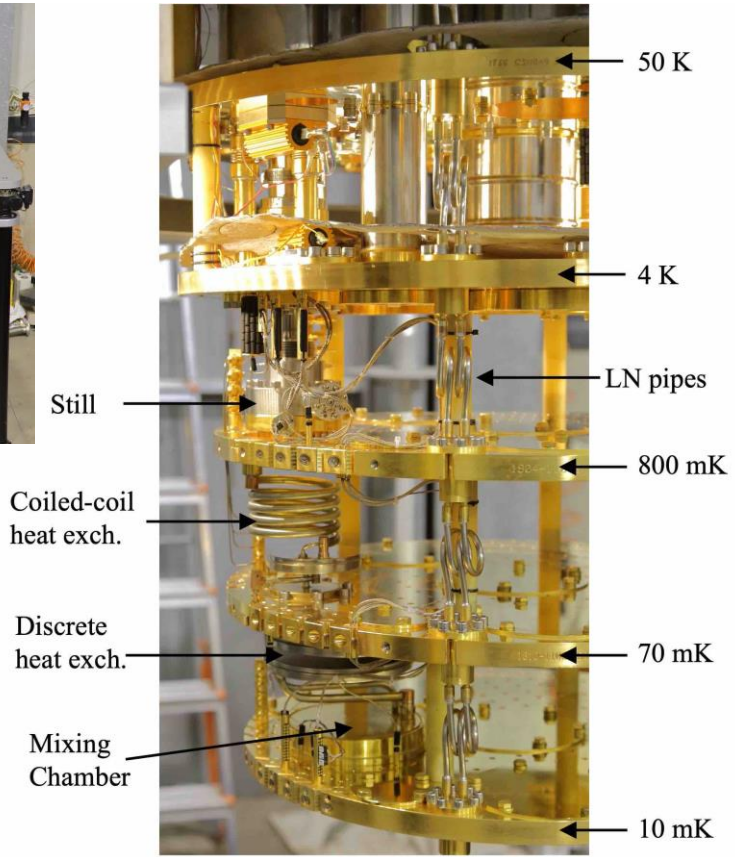
Resilience

Test the performances as a single microwave photon detector of the best device



$T_{base} = 8\ mK$

Cooling power:
$500\ \mu W\ @\ 100\ mK$





9 T magnet

- **Vast know how on Josephson effect based devices characterization**

«Stepping closer to pulsed single microwave photon detectors for axions search », *A. D'Elia et al. IEEE TAS 2022*

Research Team

Name	Unit	FTE/year	Expertise
A. D'Elia	LNF-INFN	1	JJ assembly Pre-patterned substrate design and characterization
C. Ligi	LNF-INFN	0.1	Cryogenics
J. Rezvani	University of Camerino	0.3	JJ assembly and characterization
F. Chiarello	IFN-CNR	0.15	Pre-patterned substrate production and JJ characterization
F. Mattioli	IFN-CNR	0.15	Pre-patterned substrate production and JJ characterization
		Total=1.7	

Timetable

	Months: 0-6	Months: 7-12	Months: 13-24
WP1:$NbSe_2$ flakes thickness determination			
WP2:Design of the Au pre-patterned substrate+Assembly of $NbSe_2$ JJ			
WP3:Best assembly determination and characterization of $NbSe_2$ JJ photon detection efficiency and MF resiliency			

Know-how and synergies

Resilience

Know-how

SQMS

SUPERGALAX

SIMP

DART WARS

Qub-IT

Synergies

QUÆRERE AXION



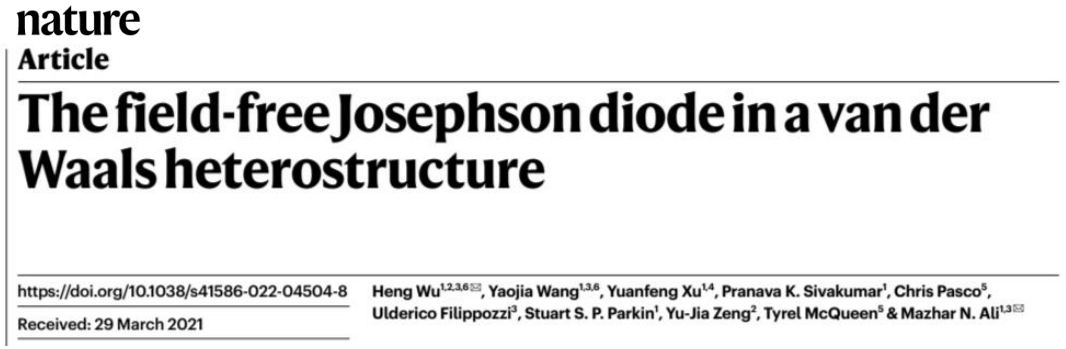
Budget

Request year 1	Typology	Single unit price [€]	Quantity	Total price [€]
2D materials assembly stage	equipment	26002	X1	26002
Vibration isolator	equipment	7100	X1	7230
Glovebox	equipment	1050.42	X1	1050.42
Sample storage with controlled atmosphere	equipment	558.76	x1	558.76
$NbSe_2$ single crystals (size ~ 1 mm)	consumable	635	X10	6350
hBN single crystal	consumable	635	X2	1270
Polymer for flakes transfer	consumable	570	X4	2280
Au powder for contacts evaporation (5 g)	consumable	1664	X2	3328
Al powder for contacts evaporation (100 g)	consumable	203	X1	203
Ti powder for contacts evaporation (100 g)	consumable	512	X1	512
Si substrates 10 x 10 mm (for patterning)	consumable	65.35	X20	1307
Si substrate 15x15mm Prepatterned standard interdigitated Au contacts	consumable	1171	X1	1171
Liquid He	consumable	17€/l	200 l	3400
Gel Pack films	consumables	8.5	x25	212.5
Electronic resist set	consumable	3397,66	x1	3397.66
Travel	Travel	About 200€/day	x30	6000
				Total= 64272.34 €

Request year 2	Typology	Single unit price [€]	Quantity	Total price [€]
$NbSe_2$ single crystals (size ~ 1 mm)	consumable	635	X10	6350
hBN single crystal	consumable	635	X2	1270
Polymer for flakes transfer	consumable	570	X4	2280
Au powder for contacts evaporation (5 g)	consumable	1664	X2	3328
Al powder for contacts evaporation (100 g)	consumable	203	X1	203
Ti powder for contacts evaporation (100 g)	consumable	512	X1	512
Si substrates 10 x 10 mm (for patterning)	consumable	65.35	X20	1307
Si substrate 15x15mm Prepatterned standard interdigitated Au contacts	consumable	1171	X1	1171
Liquid He	consumable	17€/l	200 l	3400
Gel Pack films	consumable	8.5	x25	212.5
Electronic resist set	consumable	3397,66	x1	3397.66
Travel	Travel	About 200€/day	X30	6000
				Total= 29431.16 €

Conclusions and perspectives

- RESILIENCE aims to fabricate and test a magnetic field resilient microwave single photon detector.
- To achieve this goal, we plan to assembly JJ using the Ising superconductor $NbSe_2$. In this way the JJ inherits the magnetic field resistance of the $NbSe_2$.
- Optimization of the magnetic field resiliency, through the optimization of the assembly parameters, is crucial. We will test the most promising device as a single photon detector , down to 8 mK and in presence of 9T magnetic field
- vdW materials are the perfect platform for quantum technologies and they are being more and more used to fabricate quantum circuits. Developing the know-how and the instrumentations necessary to build all-vdW quantum components will be extremely rewarding for the future of INFN in the field of quantum technologies


nature

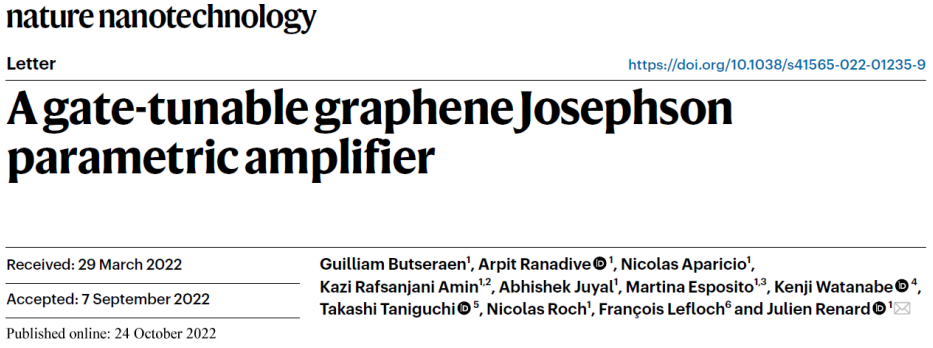
Article

The field-free Josephson diode in a van der Waals heterostructure

https://doi.org/10.1038/s41586-022-04504-8

Received: 29 March 2021

Heng Wu[1,2,3,6], Yaojia Wang[1,3,6], Yuanfeng Xu[1,4], Pranava K. Sivakumar[1], Chris Pasco[5], Ulderico Filippozzi[3], Stuart S. P. Parkin[1], Yu-Jia Zeng[2], Tyrel McQueen[5] & Mazhar N. Ali[1,3]


nature nanotechnology

Letter

https://doi.org/10.1038/s41565-022-01235-9

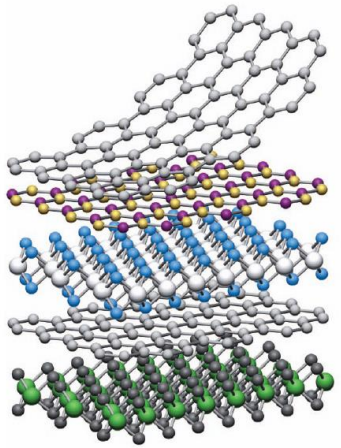
A gate-tunable graphene Josephson parametric amplifier

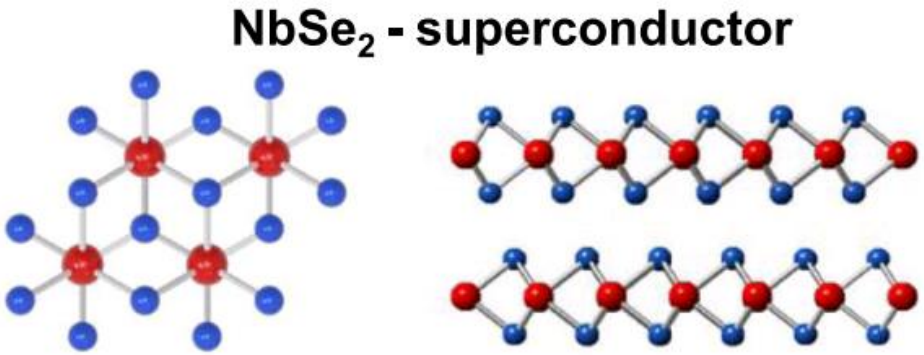
Received: 29 March 2022

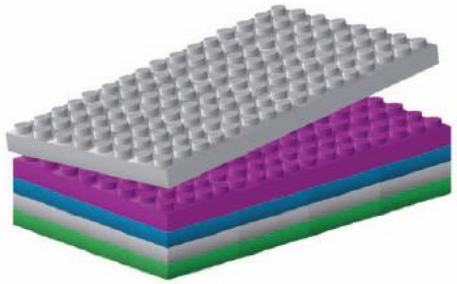
Accepted: 7 September 2022

Published online: 24 October 2022

Guilliam Butseraen[1], Arpit Ranadive[1], Nicolas Aparicio[1], Kazi Rafsanjani Amin[1,2], Abhishek Juyal[1], Martina Esposito[1,3], Kenji Watanabe[4], Takashi Taniguchi[5], Nicolas Roch[1], François Lefloch[6] and Julien Renard[1]







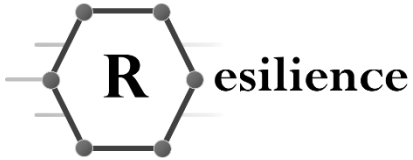
Thank you for your attention!

Resilience

INFN
Istituto Nazionale di Fisica Nucleare

COLD
LNF
CryOgenic Laboratory for Detectors

Backup slides



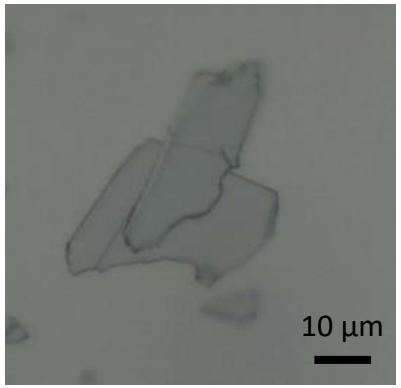
Axions conversion power

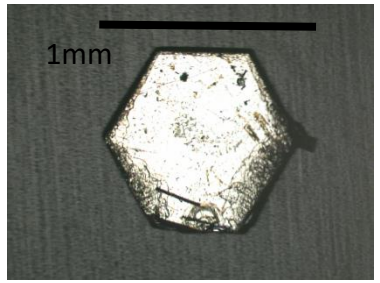
Resilience

$$P_a = \left(\frac{g_{a\gamma\gamma}^2}{m_a^2} \hbar^3 c^3 \rho_a \right) \left(\frac{\beta}{1+\beta} \omega_c \frac{1}{\mu_0} B_0^2 V C_{030} Q_L \right) \left(\frac{1}{1 + (2Q_L \, \Delta_\omega / \omega_c)^2} \right)$$

Risk assessement

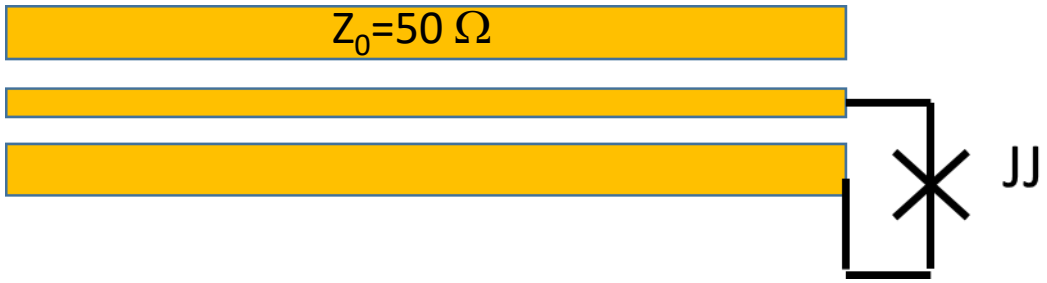
Risk	Consequences and solutions
Missing delivery or malfunctioning of the vdW flake assembly station	This event cannot critically slow down RESILIENCE realization. We can produce $NbSe_2$ junctions @Univeristy of Camerino, in sub-optimal conditions. Feasibility test junction displayed on the right.
Missing delivery or delayed delivery of the $NbSe_2$ crystals.	We already possess a small amount of $NbSe_2$, we estimate that it will be enough for the first 3-4 months where the usage of of $NbSe_2$ is not yet maximum since the JJ assembly will start after flakes thickness calibration. On the right a singe crystal of $NbSe_{2.+}$ Thanks to Prof. Changyoung Kim from Seoul University for the single crystals specimens supply
Unavailability of electrodes patterning equipment	In case the lithography equipment necessary to fabricate the chips with pre-patterned Au contacts is unavailable (for any reason), RESILIENCE would still work but in sub-optimal condition. In the short term, it is possible to buy standard Si chips with interdigitated Au pre patterned contacts that can replace the custom chip produced by the IFN-CNR





JJ coupled to a transmission line

Junction coupled to a transmission line

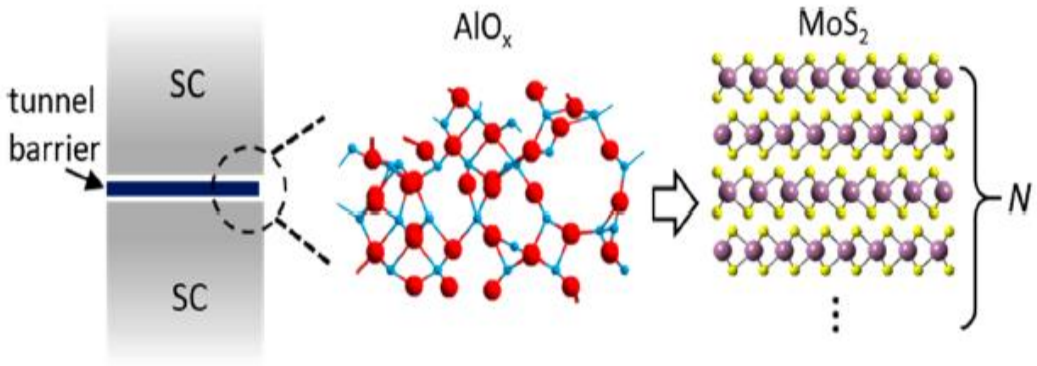


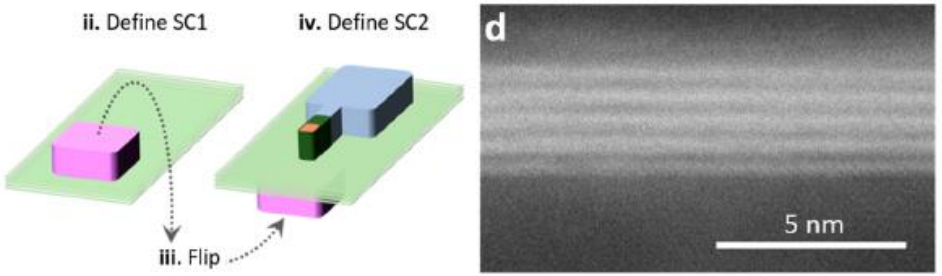
The JJ terminated on a 50 Ω ensures a high overall system efficiency. The “isolated” junction, on the other hand, has a high internal quantum efficiency, and a small overall efficiency.

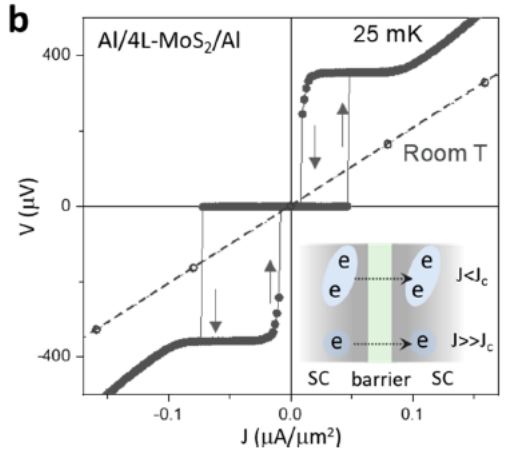
vdW integration with standard fabrication technologies

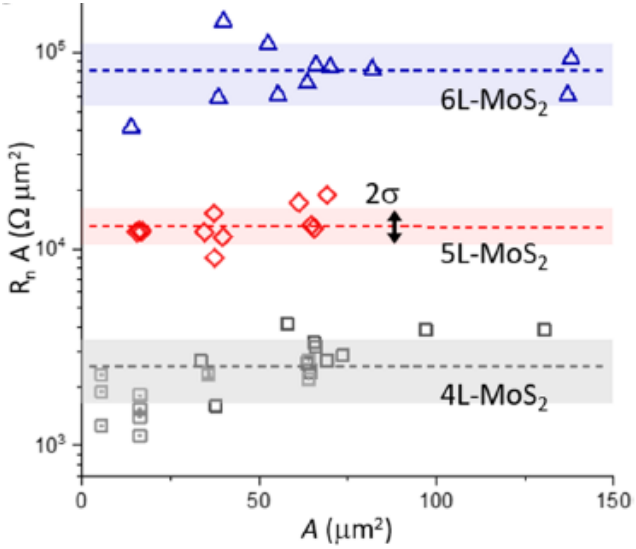
Resilience

Replace amorphous AlO_x layer with controlled number of MoS_2 layers









Maximum control over R_n therefore $I_c \propto \frac{1}{R_n}$

K. Lee et al, Nano Lett. 2019